

```
In [1]: import numpy as np
import pandas as pd
```

```
In [2]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
#plt.style.use('fivethirtyeight')
import warnings
warnings.filterwarnings('ignore')
```

```
In [3]: iris=pd.read_csv(r'C:\Users\HAI\Downloads\Iris.csv')
```

```
In [4]: iris
```

```
Out[4]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

```
In [5]: iris.head()
```

```
Out[5]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
In [6]: iris.drop('Id',axis=1,inplace=True)
```

```
In [7]: iris.head()
```

```
Out[7]:
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

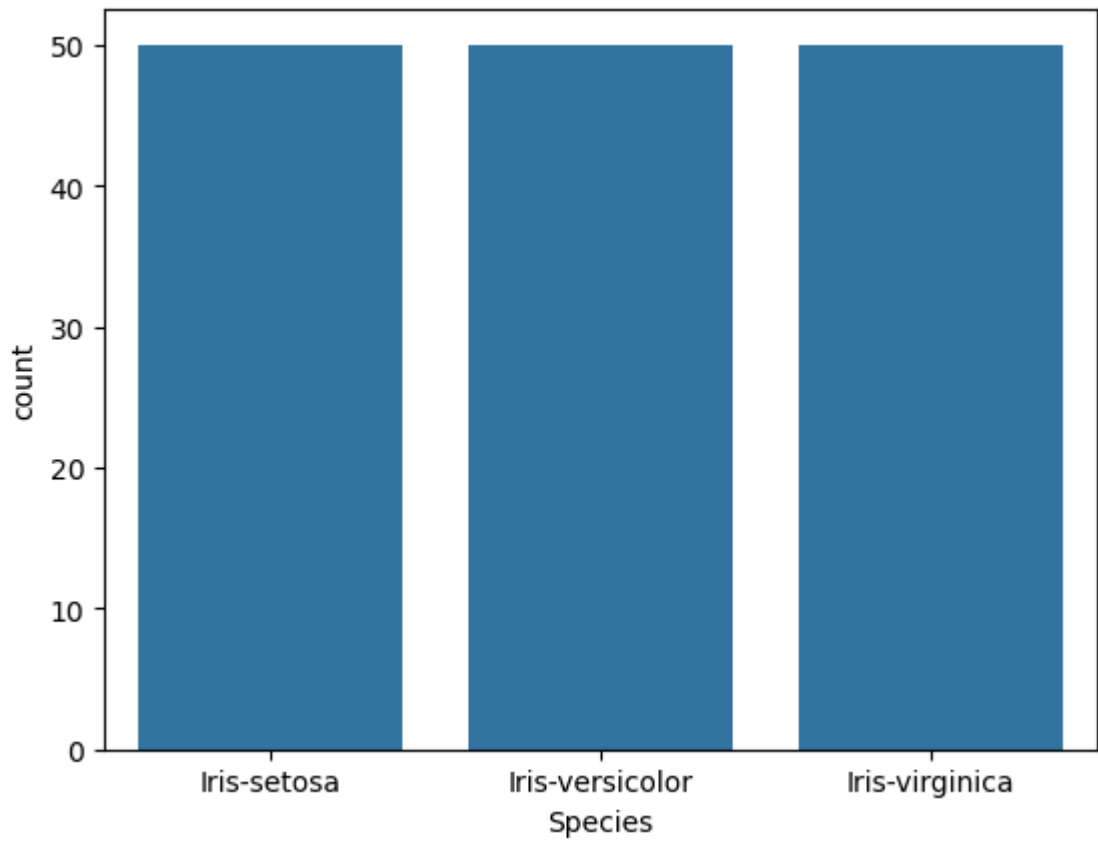
```
In [8]: iris.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   SepalLengthCm   150 non-null   float64
1   SepalWidthCm    150 non-null   float64
2   PetalLengthCm   150 non-null   float64
3   PetalWidthCm    150 non-null   float64
4   Species         150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
In [9]: iris['Species'].value_counts()
```

```
Out[9]: Species
Iris-setosa      50
Iris-versicolor  50
Iris-virginica   50
Name: count, dtype: int64
```

```
In [10]: sns.countplot(x='Species', data=iris)
plt.show()
```

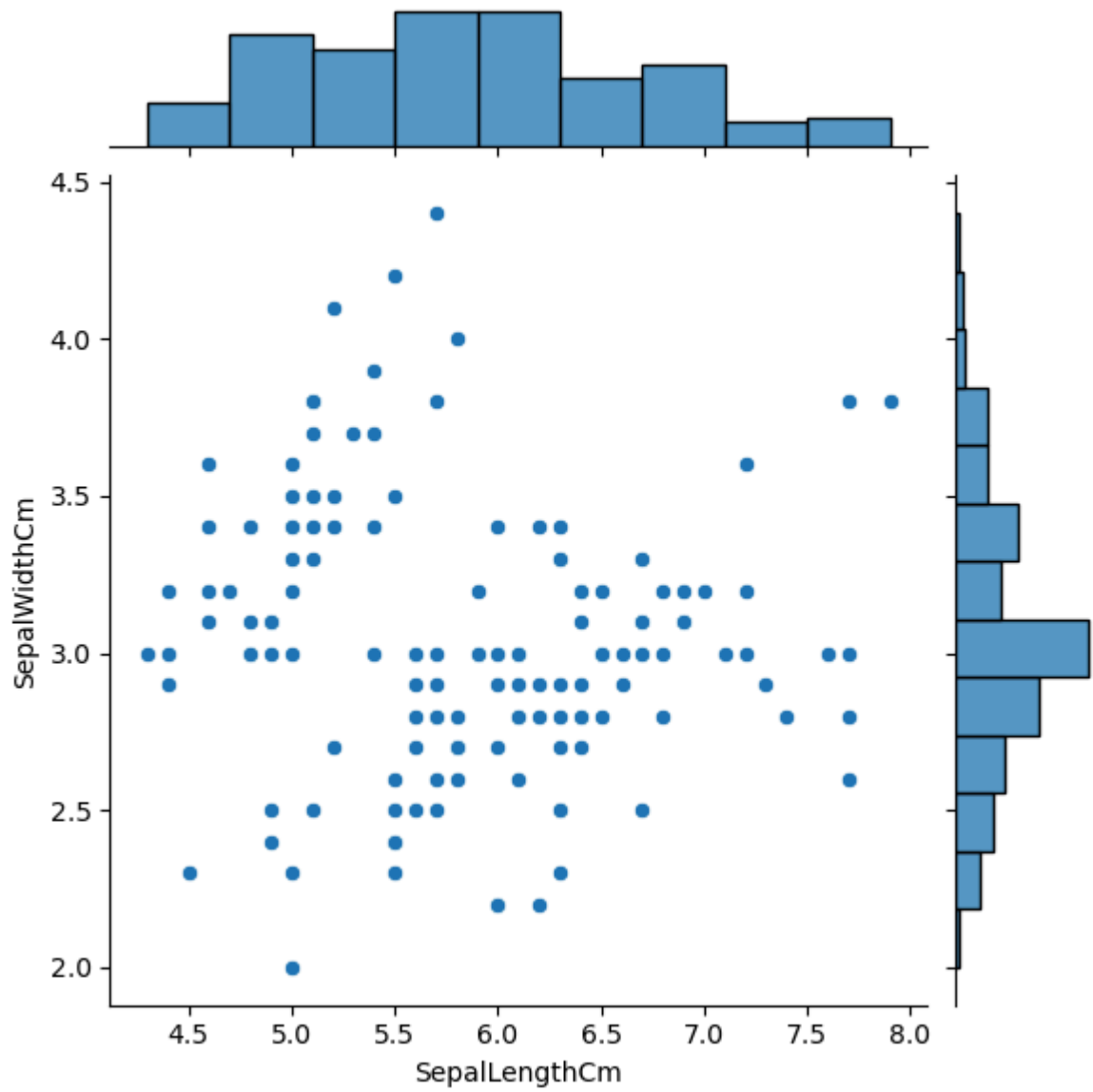


```
In [11]: iris.head()
```

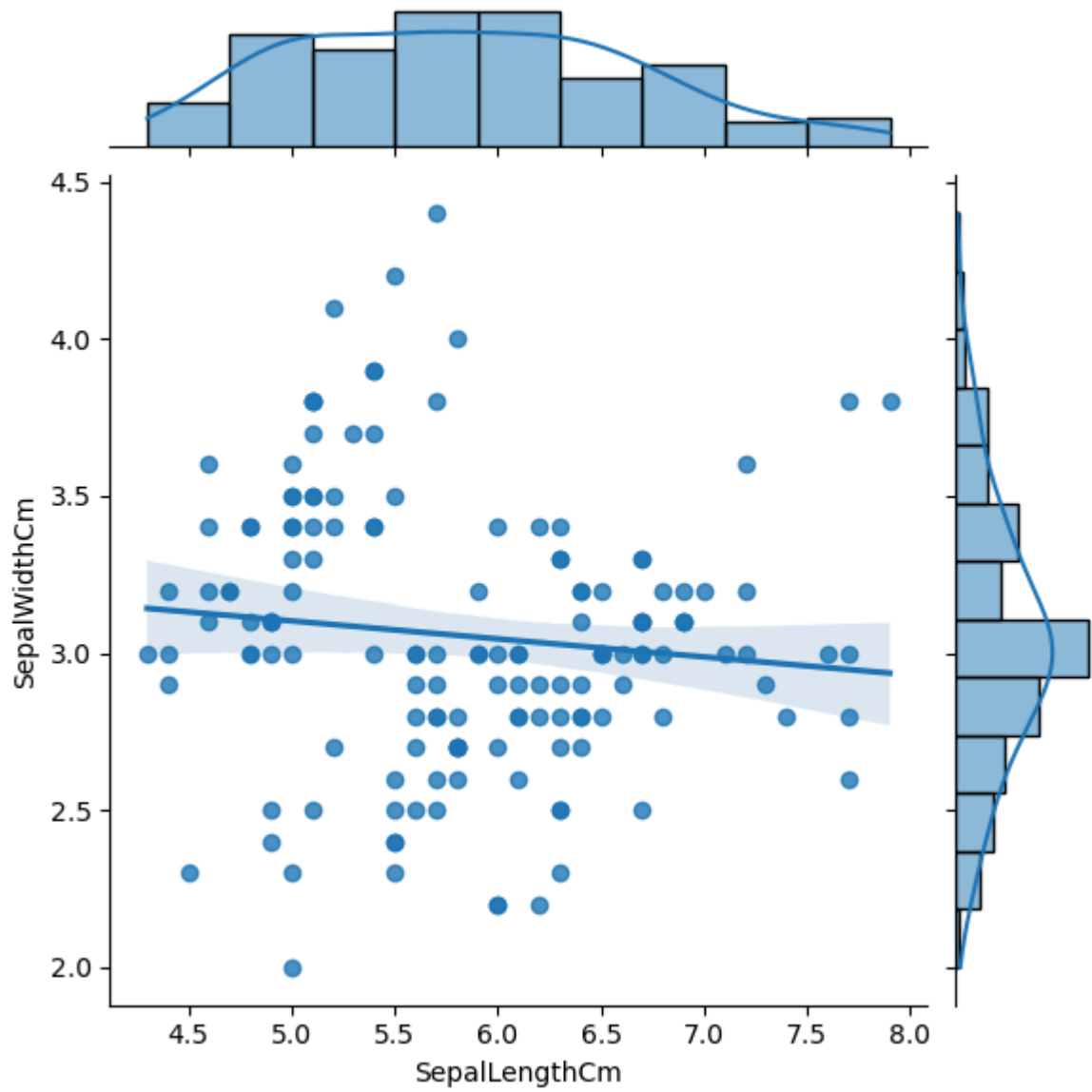
```
Out[11]:
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

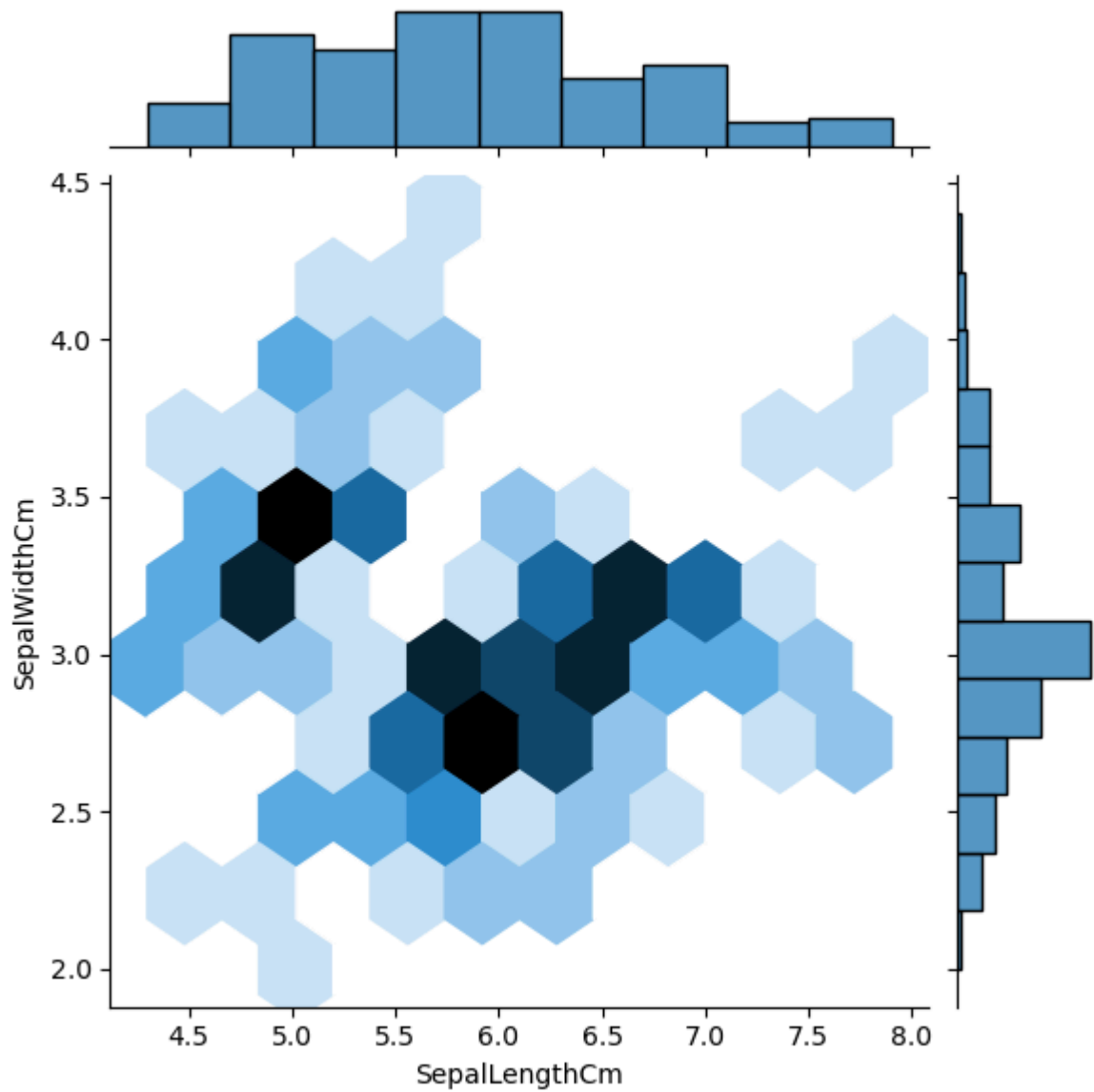
```
In [12]: fig=sns.jointplot(x='SepalLengthCm',y='SepalWidthCm',data=iris)
plt.show()
```



```
In [13]: sns.jointplot(x="SepalLengthCm", y="SepalWidthCm", data=iris, kind="reg")  
plt.show()
```



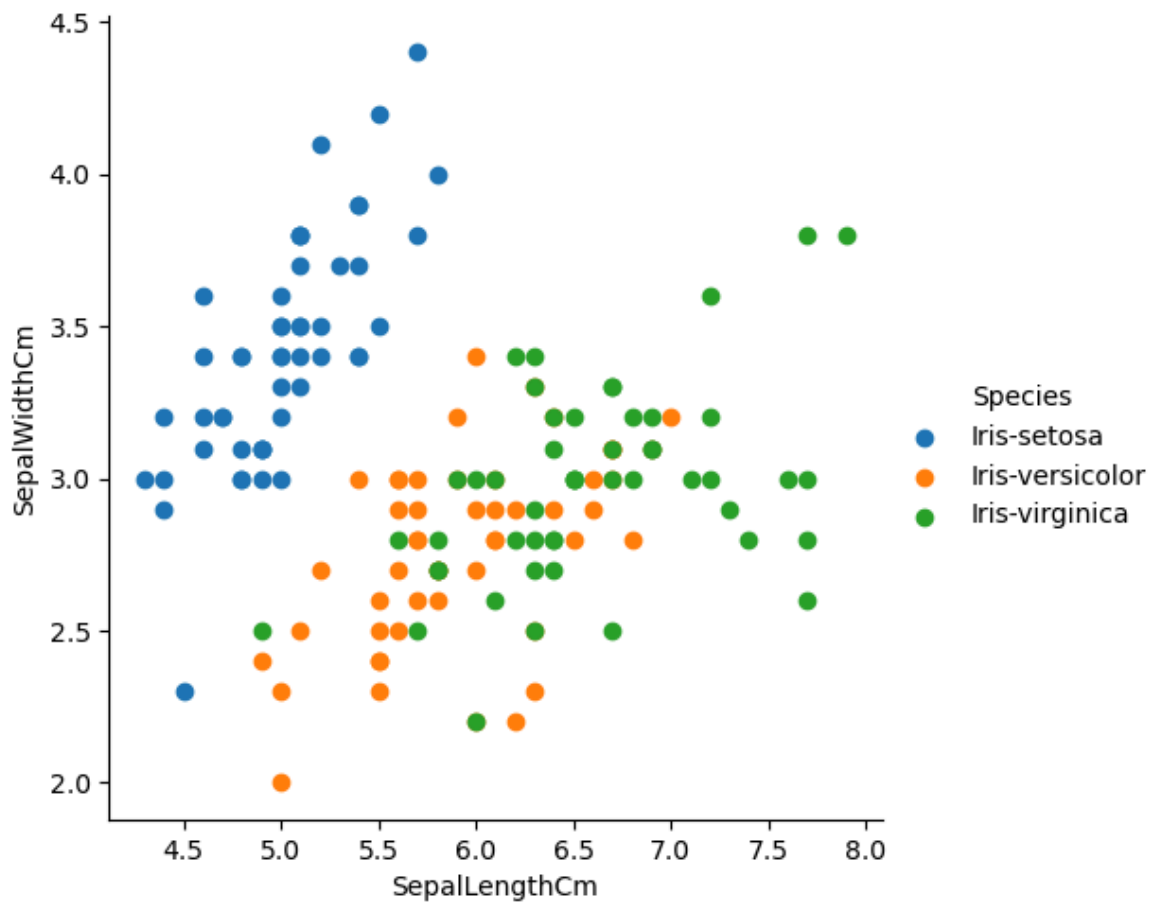
```
In [14]: fig=sns.jointplot(x='SepalLengthCm',y='SepalWidthCm',kind='hex',data=iris)
```



```
In [15]: import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline

# Using the newer argument name `height` instead of deprecated `size`
g = sns.FacetGrid(iris, hue='Species', height=5)
g.map(plt.scatter, 'SepalLengthCm', 'SepalWidthCm') \
    .add_legend()

plt.show()
```



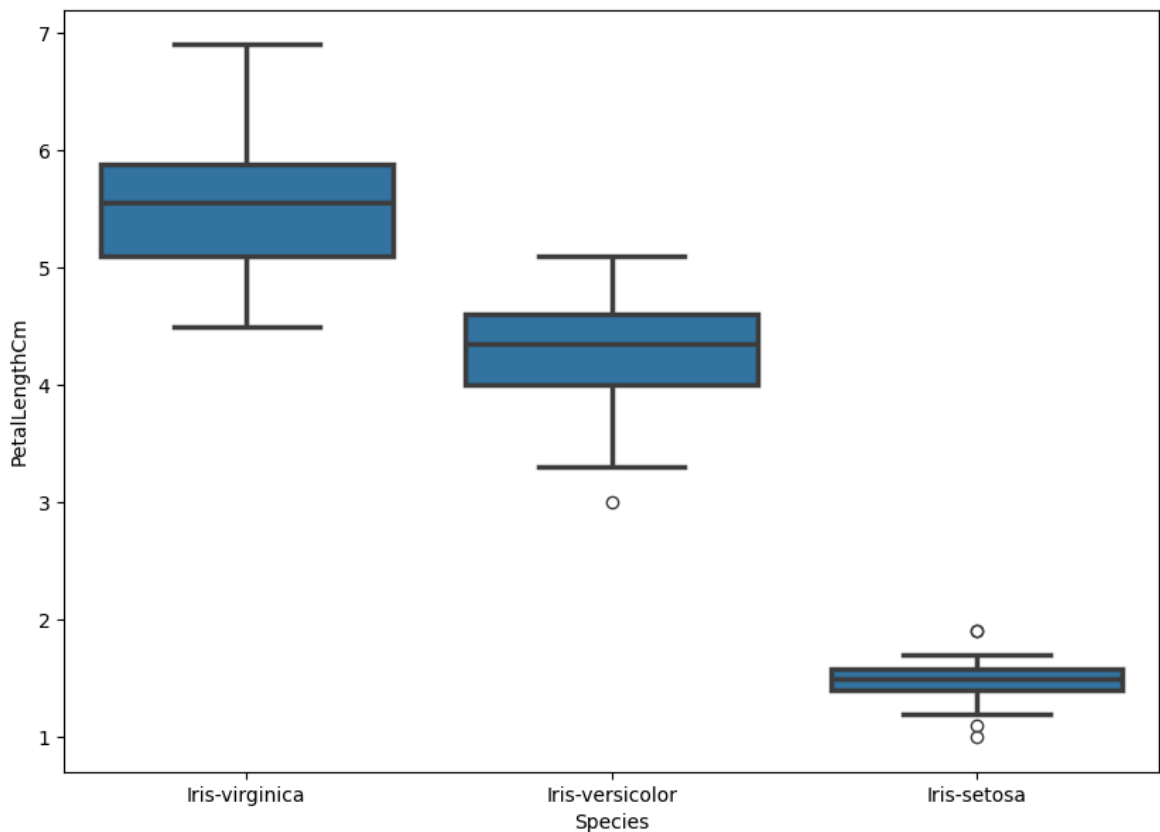
```
In [16]: iris.head()
```

```
Out[16]:
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

```
In [17]: fig=plt.gcf()
fig.set_size_inches(10,7)
fig=sns.boxplot(x='Species',y='PetalLengthCm',data=iris,order=['Iris-virginica',
```

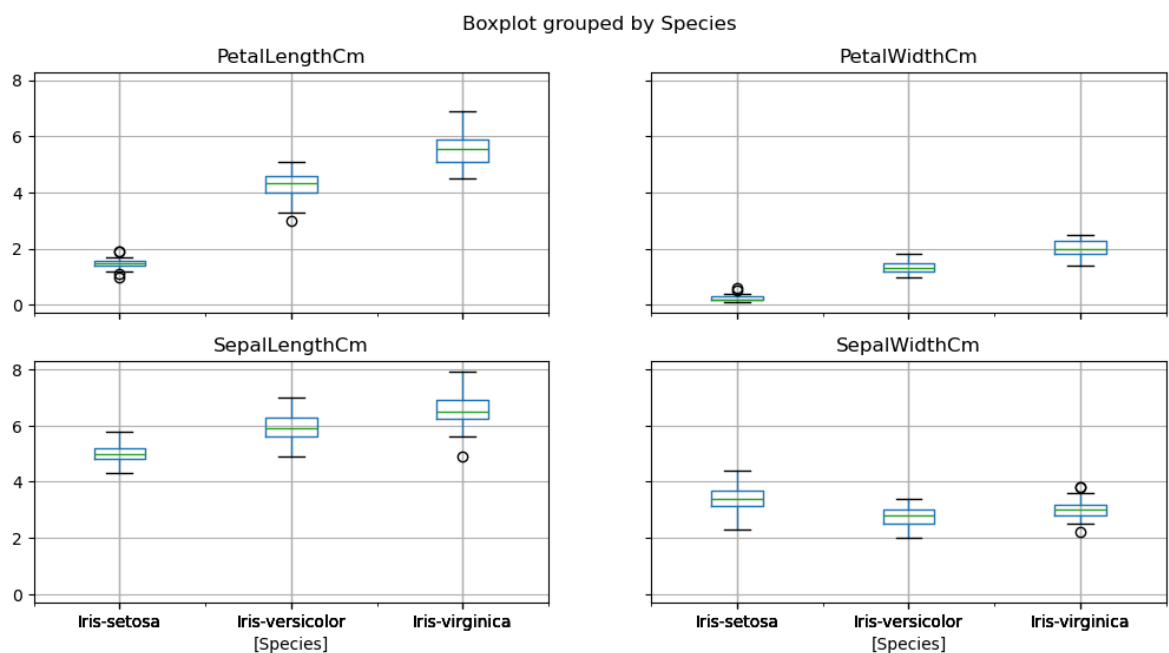
```
In [18]: plt.show()
```



```
In [19]: #iris.drop("Id", axis=1).boxplot(by="Species", figsize=(12, 6))
iris.boxplot(by="Species", figsize=(12, 6))
```

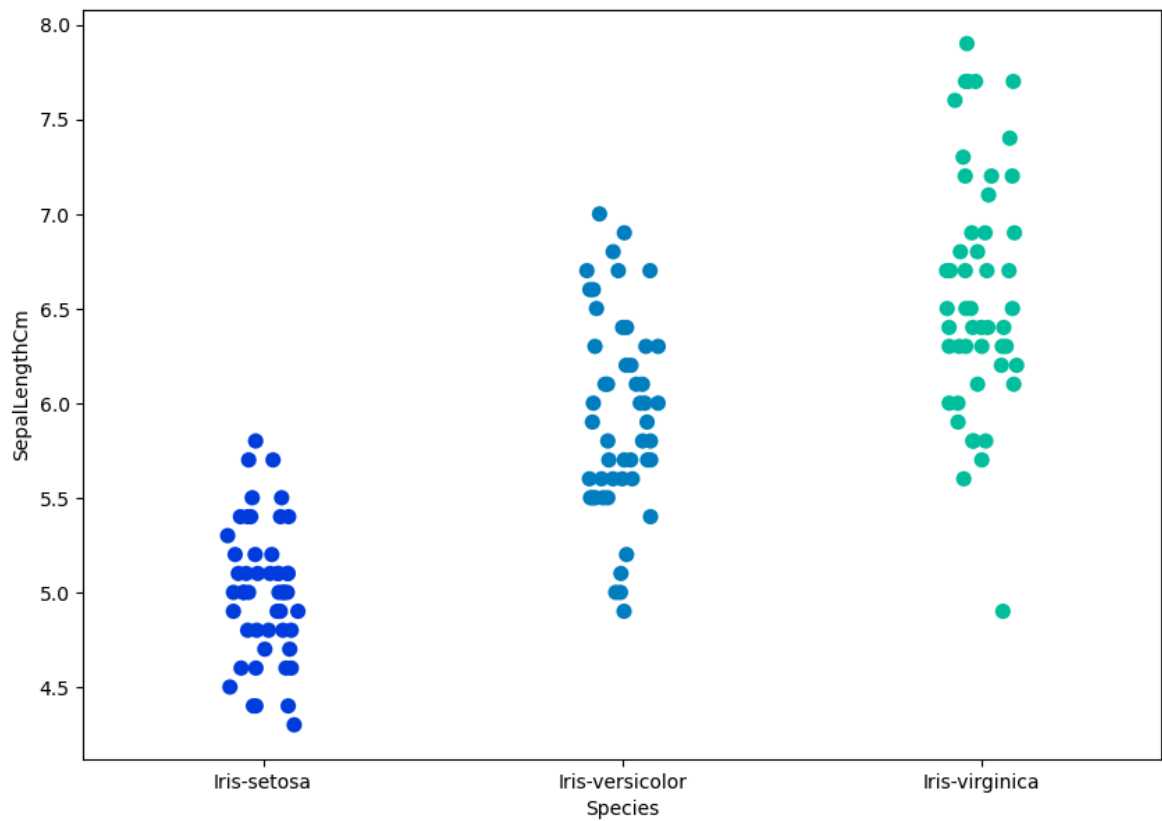
```
Out[19]: array([[<Axes: title={'center': 'PetalLengthCm'}, xlabel='[Species]'],
  <Axes: title={'center': 'PetalWidthCm'}, xlabel='[Species]'],
  [<Axes: title={'center': 'SepalLengthCm'}, xlabel='[Species]'],
  <Axes: title={'center': 'SepalWidthCm'}, xlabel='[Species]']],
  dtype=object)
```

```
In [20]: plt.show()
```

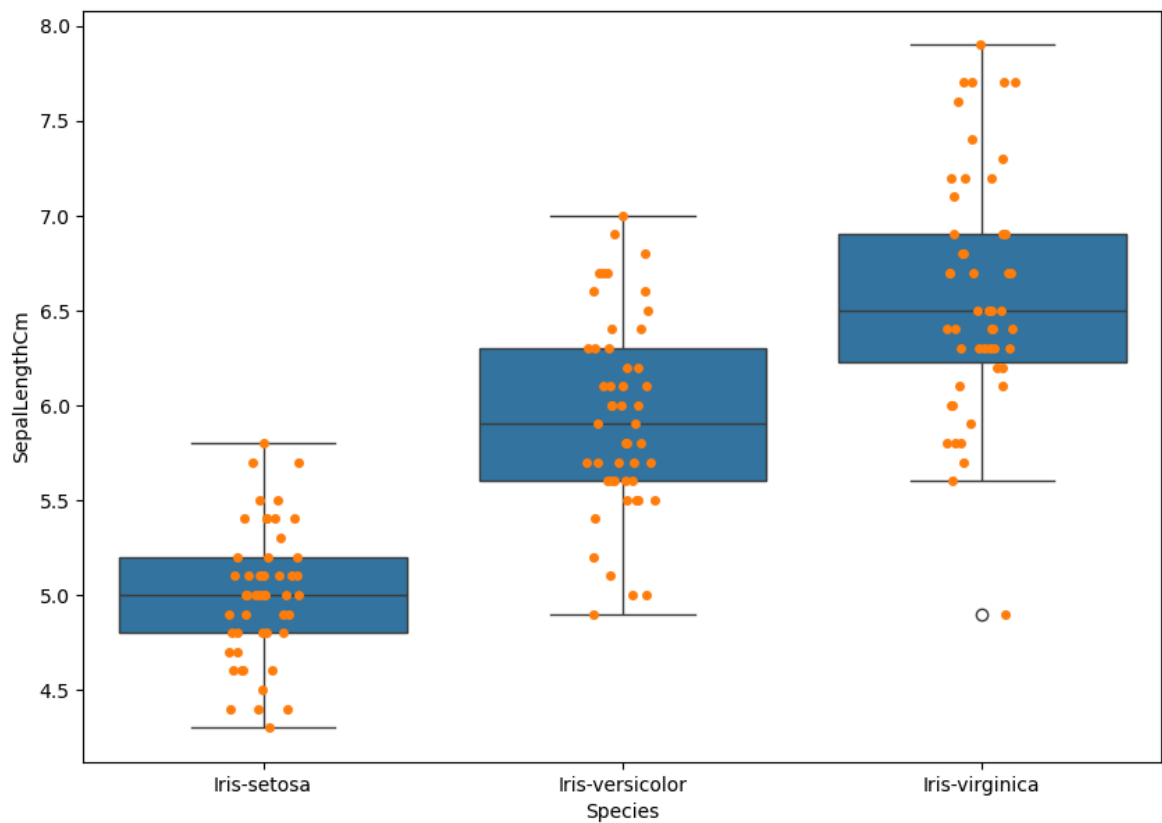


```
In [21]: fig=plt.gcf()
fig.set_size_inches(10,7)
```

```
fig=sns.stripplot(x='Species',y='SepalLengthCm',data=iris,jitter=True,edgecolor=
plt.show()
```



```
In [22]: fig=plt.gcf()
fig.set_size_inches(10,7)
fig=sns.boxplot(x='Species',y='SepalLengthCm',data=iris)
fig=sns.stripplot(x='Species',y='SepalLengthCm',data=iris,jitter=True,edgecolor=
plt.show()
```



```
In [23]: ax = sns.boxplot(x="Species", y="PetallLengthCm", data=iris)
ax = sns.stripplot(x="Species", y="PetallLengthCm", data=iris, jitter=True, edgecolor=

boxtwo = ax.artists[2]
boxtwo.set_facecolor('yellow')
boxtwo.set_edgecolor('black')
boxthree=ax.artists[1]
boxthree.set_facecolor('red')
boxthree.set_edgecolor('black')
boxthree=ax.artists[0]
boxthree.set_facecolor('green')
boxthree.set_edgecolor('black')

plt.show()
```

```
-----
IndexError                                Traceback (most recent call last)
Cell In[23], line 4
      1 ax= sns.boxplot(x="Species", y="PetallLengthCm", data=iris)
      2 ax= sns.stripplot(x="Species", y="PetallLengthCm", data=iris, jitter=True,
edgecolor="gray")
----> 4 boxtwo = ax.artists[2]
      5 boxtwo.set_facecolor('yellow')
      6 boxtwo.set_edgecolor('black')

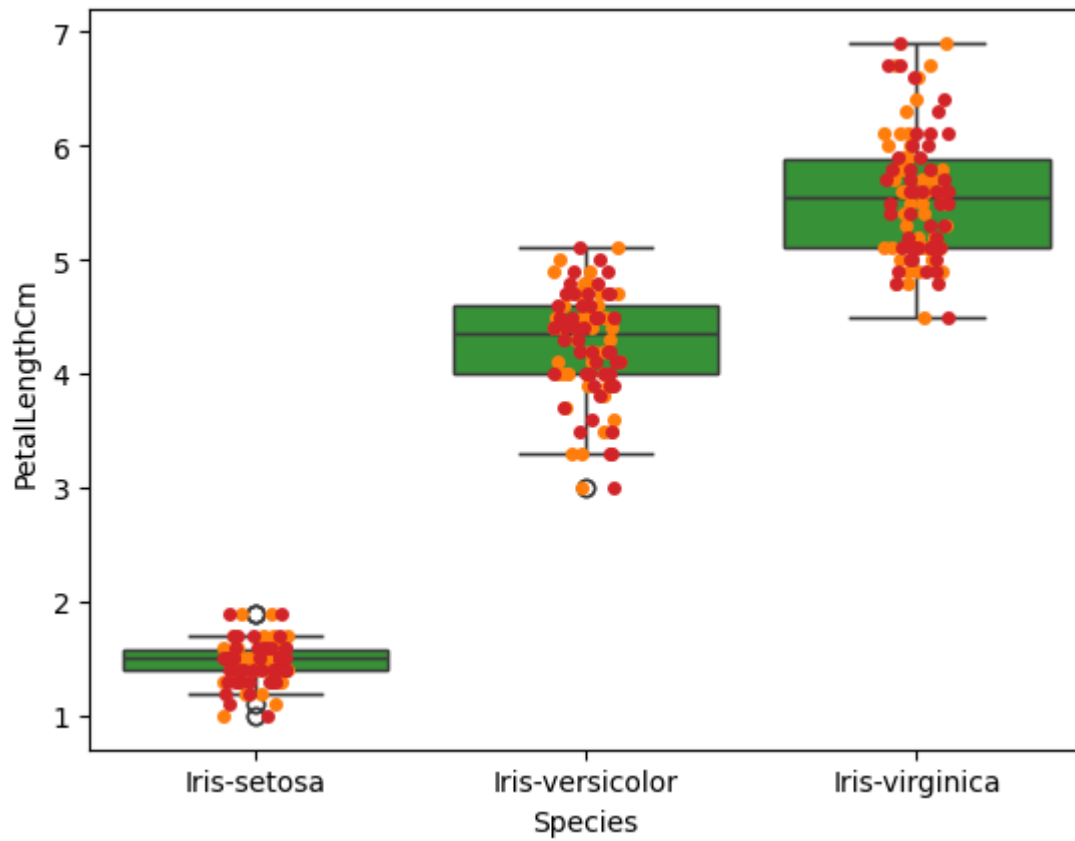
File ~\anaconda3\Lib\site-packages\matplotlib\axes\_base.py:1453, in _AxesBase.ArtistList.__getitem__(self, key)
    1452 def __getitem__(self, key):
-> 1453     return [artist
    1454               for artist in self._axes._children
    1455               if self._type_check(artist)][key]

IndexError: list index out of range
```

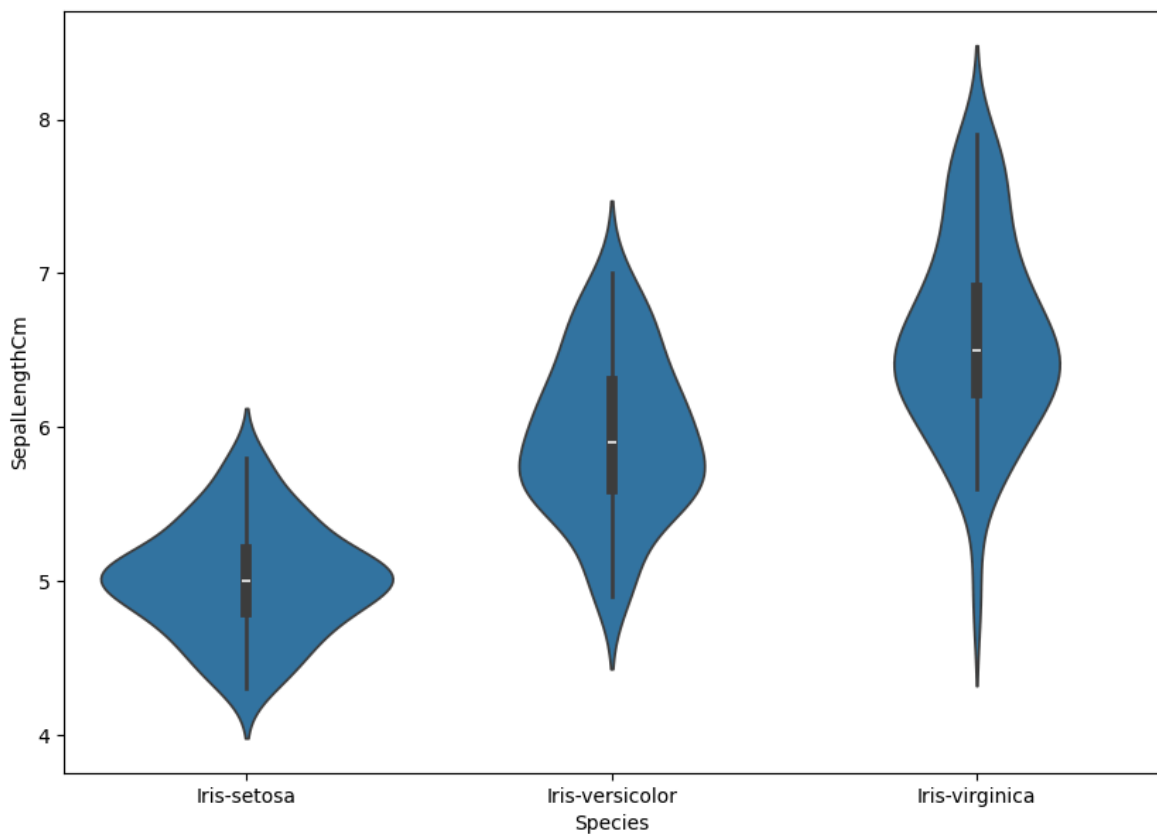
```
In [24]: # Boxplot + Stripplot
ax = sns.boxplot(x="Species", y="PetallLengthCm", data=iris)
sns.stripplot(x="Species", y="PetallLengthCm", data=iris, jitter=True, edgecolor=

# Customizing individual box colors
colors = ['green', 'red', 'yellow']
for i, box in enumerate(ax.artists):
    box.set_facecolor(colors[i])
    box.set_edgecolor('black')

plt.show()
```

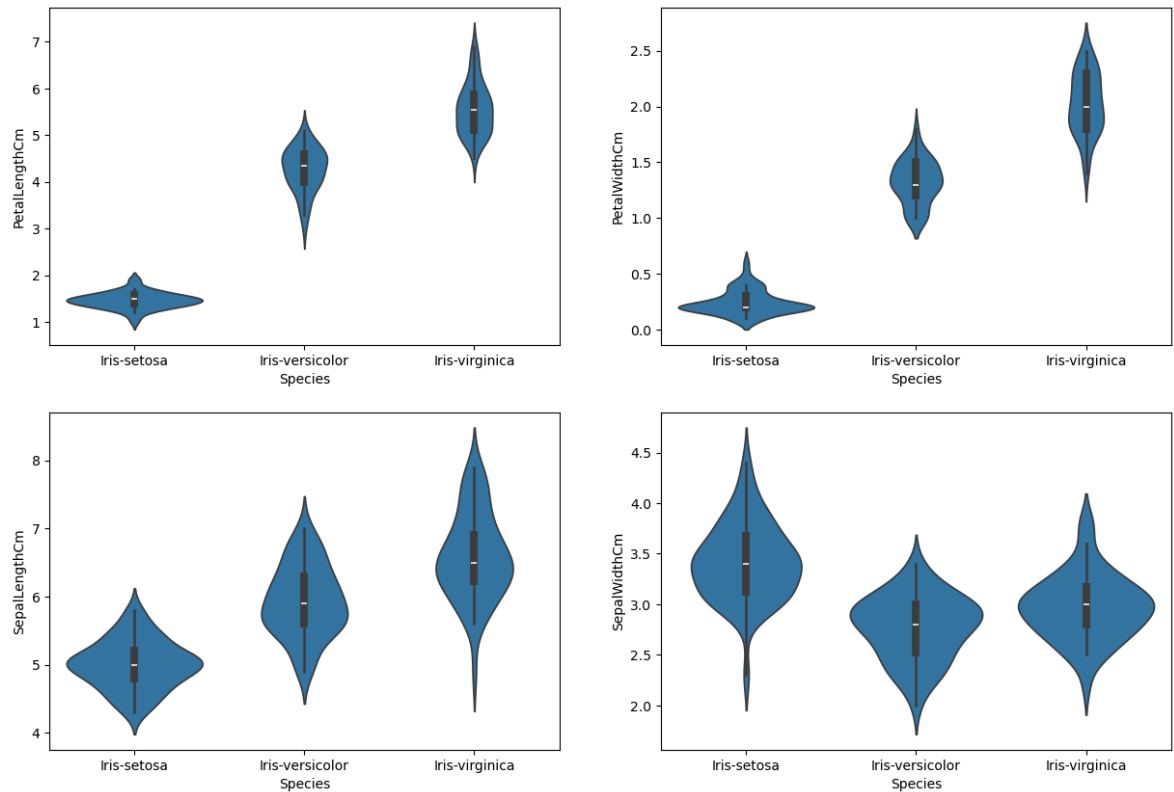


```
In [25]: fig=plt.gcf()
fig.set_size_inches(10,7)
fig=sns.violinplot(x='Species',y='SepalLengthCm',data=iris)
plt.show()
```

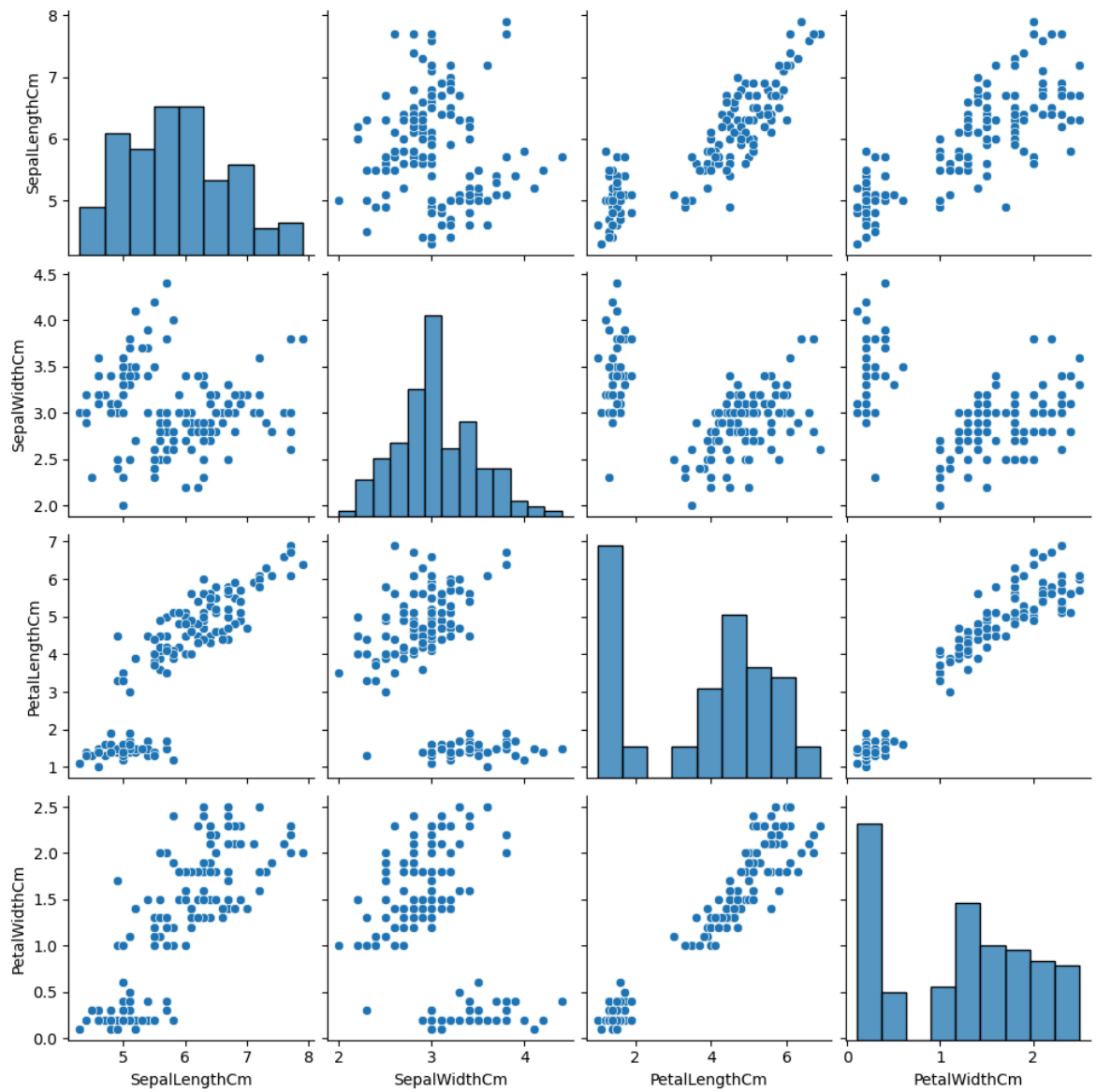


```
In [26]: plt.figure(figsize=(15,10))
plt.subplot(2,2,1)
sns.violinplot(x='Species',y='PetalLengthCm',data=iris)
```

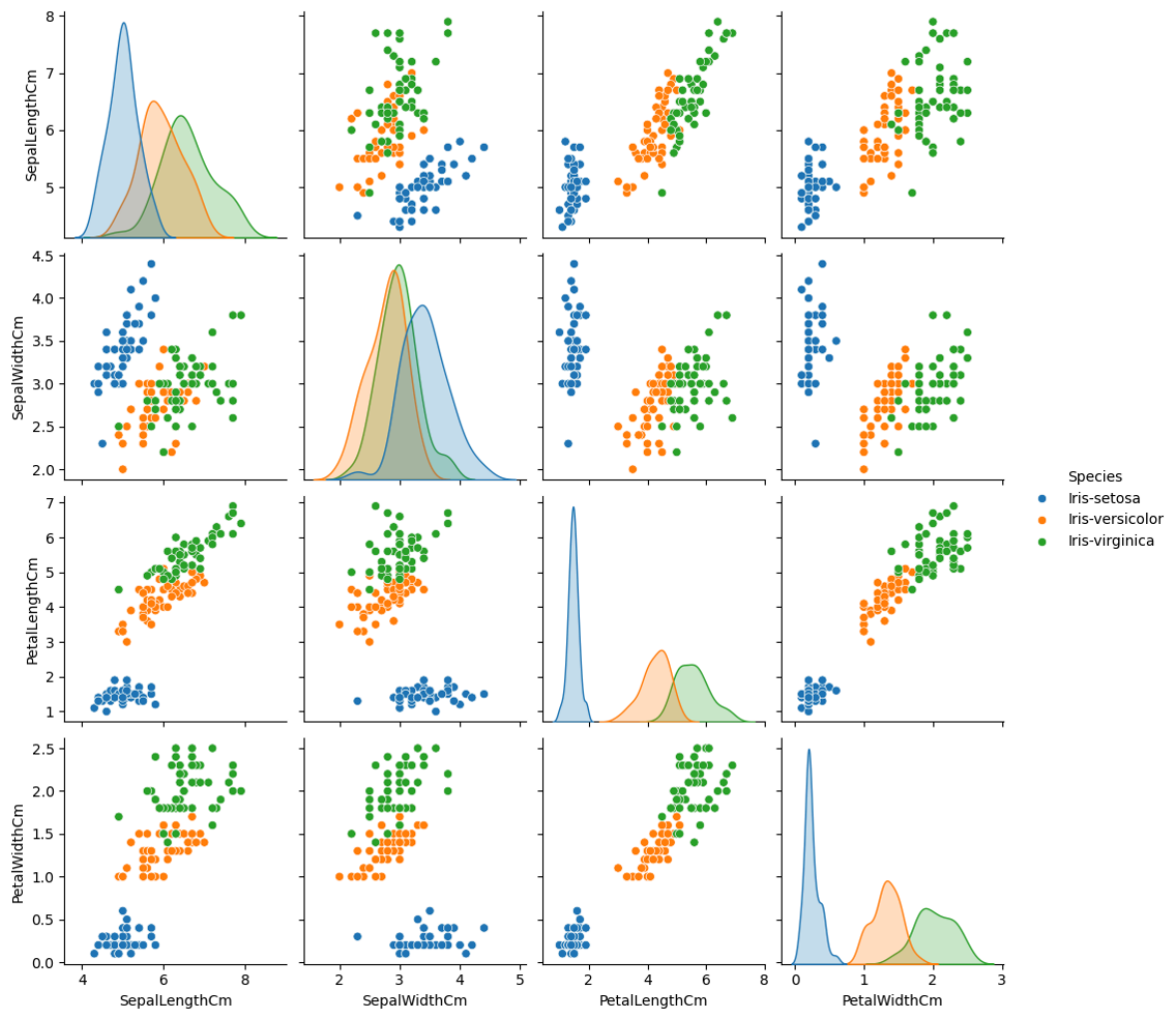
```
plt.subplot(2,2,2)
sns.violinplot(x='Species',y='PetalWidthCm',data=iris)
plt.subplot(2,2,3)
sns.violinplot(x='Species',y='SepalLengthCm',data=iris)
plt.subplot(2,2,4)
sns.violinplot(x='Species',y='SepalWidthCm',data=iris)
plt.show()
```



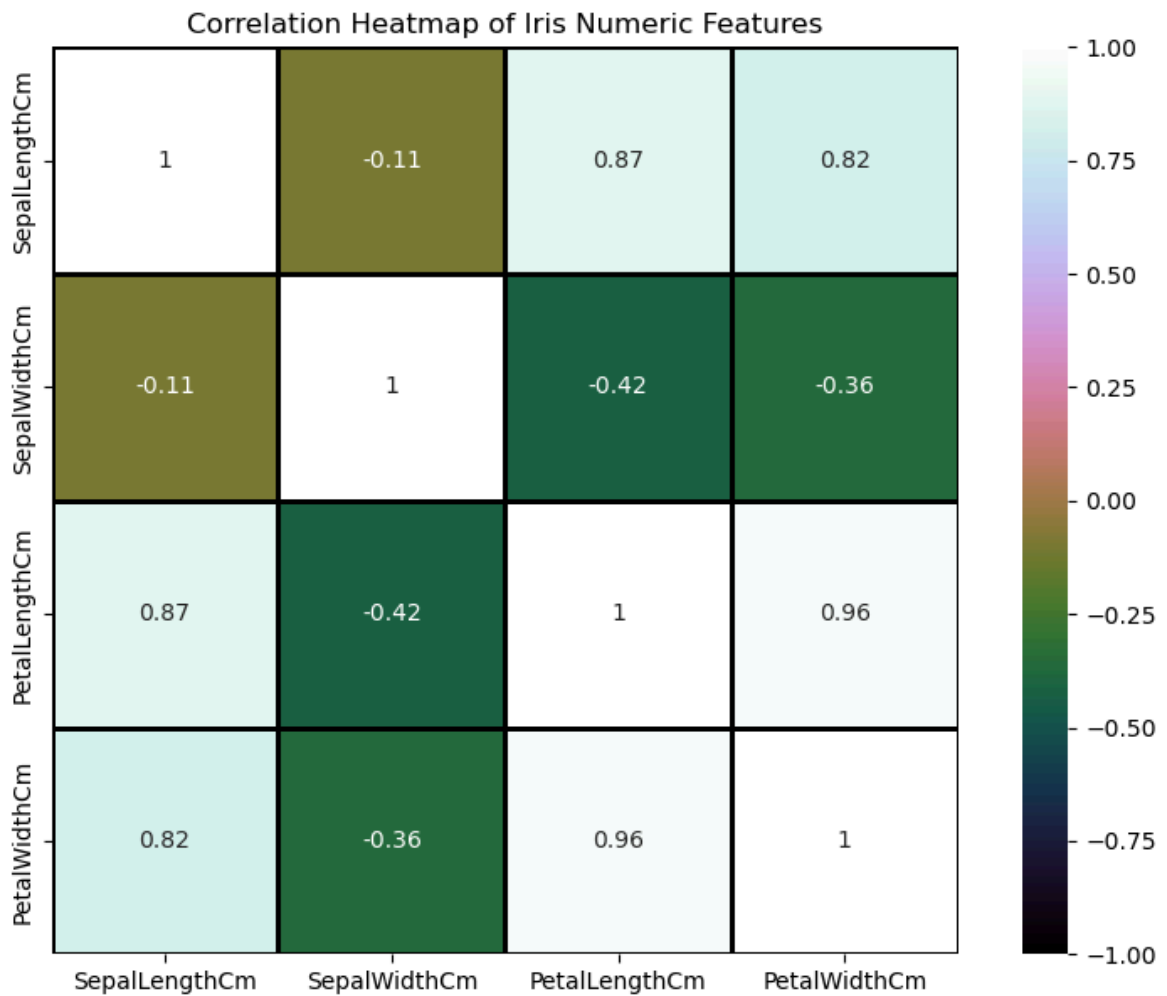
```
In [27]: sns.pairplot(data=iris,kind='scatter')
plt.show()
```



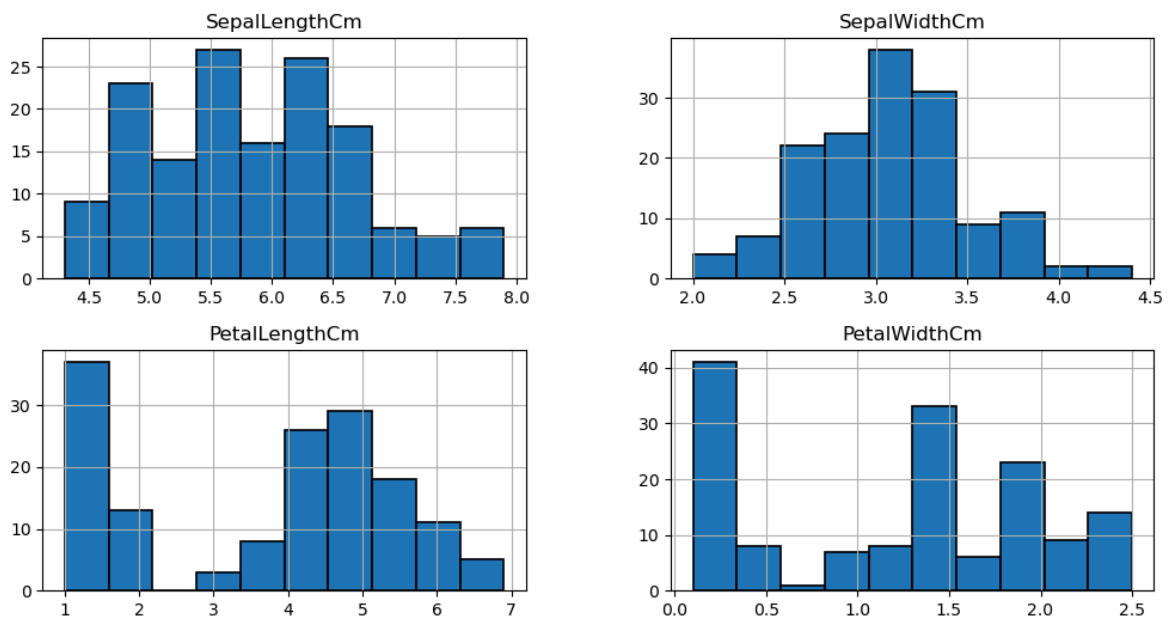
```
In [28]: sns.pairplot(iris,hue='Species');  
plt.show()
```



```
In [29]: import seaborn as sns
import matplotlib.pyplot as plt
# Drop non-numeric column(s)
iris_numeric = iris.drop(columns=['Species'])
# Set figure size
plt.figure(figsize=(10, 7))
# Plot heatmap
sns.heatmap(
    iris_numeric.corr(),
    annot=True,
    cmap='cubehelix',
    linewidths=1,
    linecolor='k',
    square=True,
    vmin=-1,
    vmax=1,
    cbar_kws={"orientation": "vertical"},
    cbar=True
)
plt.title("Correlation Heatmap of Iris Numeric Features")
plt.show()
```

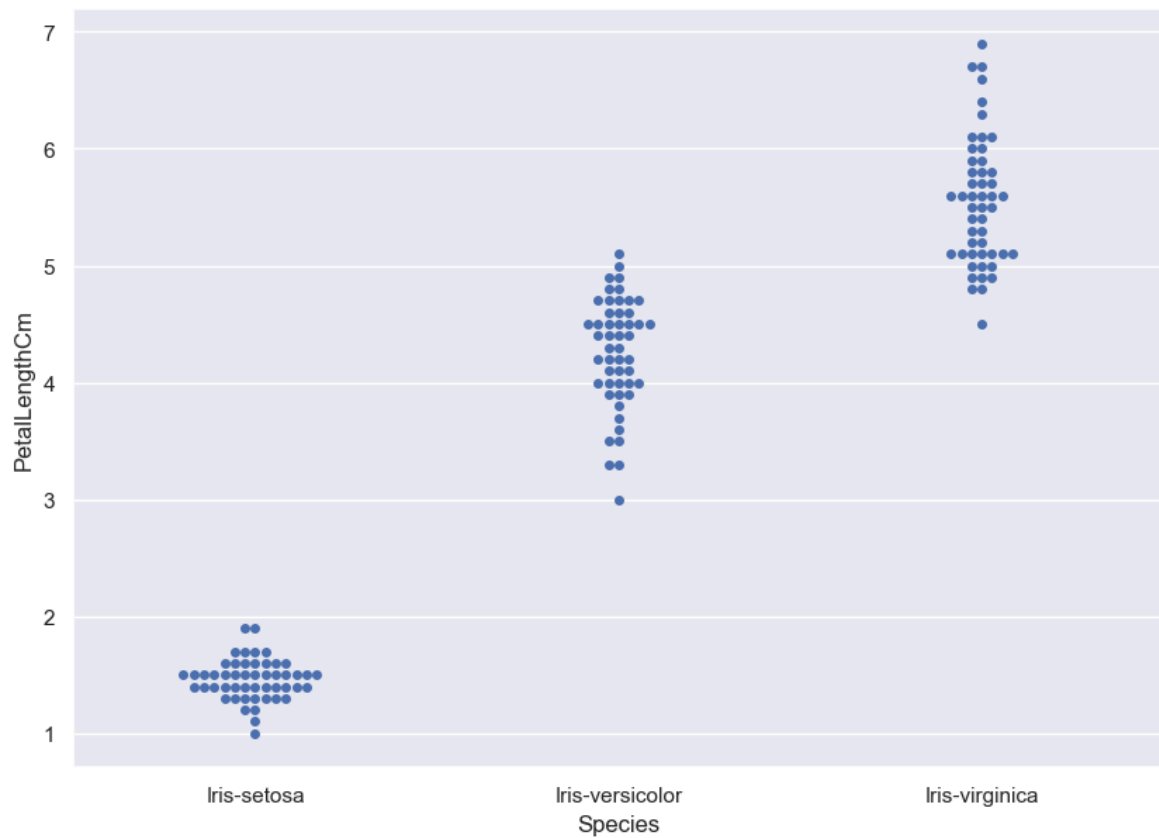


```
In [30]: iris.hist(edgecolor='black', linewidth=1.2)
fig = plt.gcf() # Get current figure
fig.set_size_inches(12, 6) # Corrected 'fig' instead of 'ig'
plt.show()
```

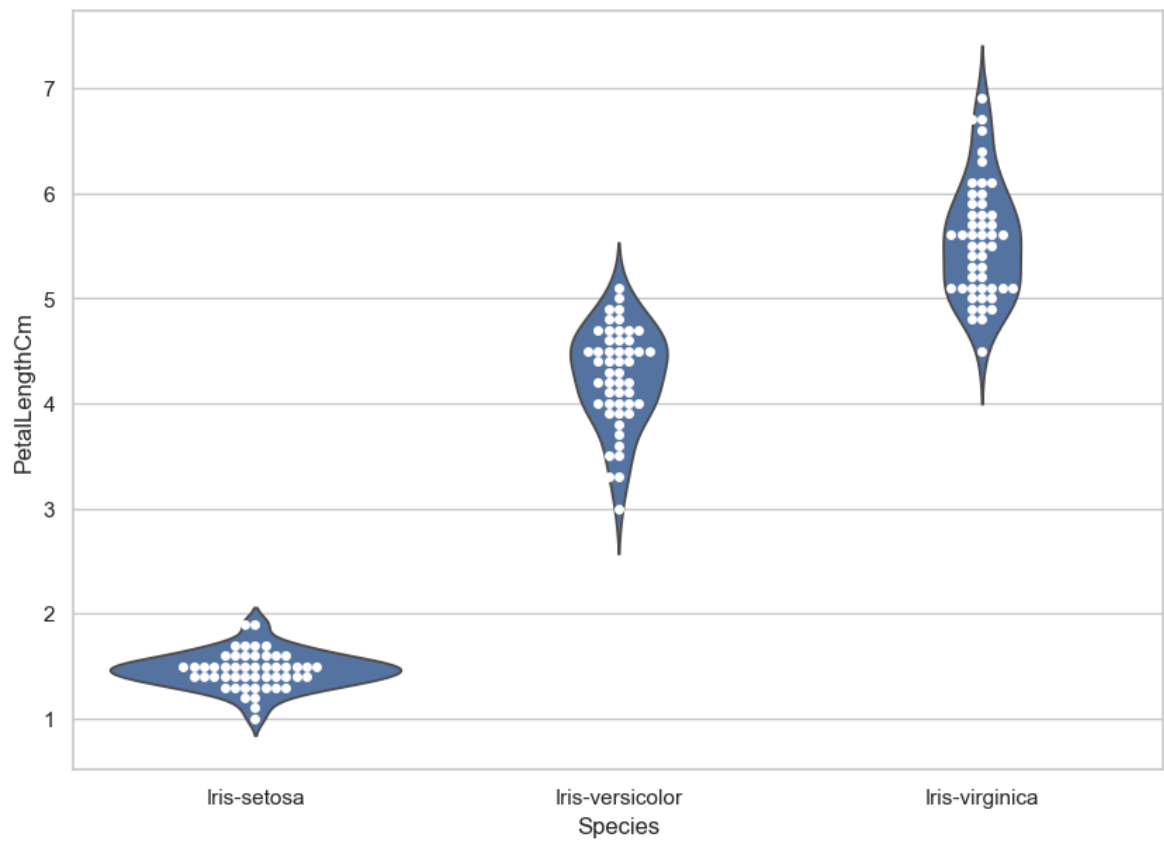


```
In [31]: sns.set(style="darkgrid")
fig=plt.gcf()
fig.set_size_inches(10,7)
```

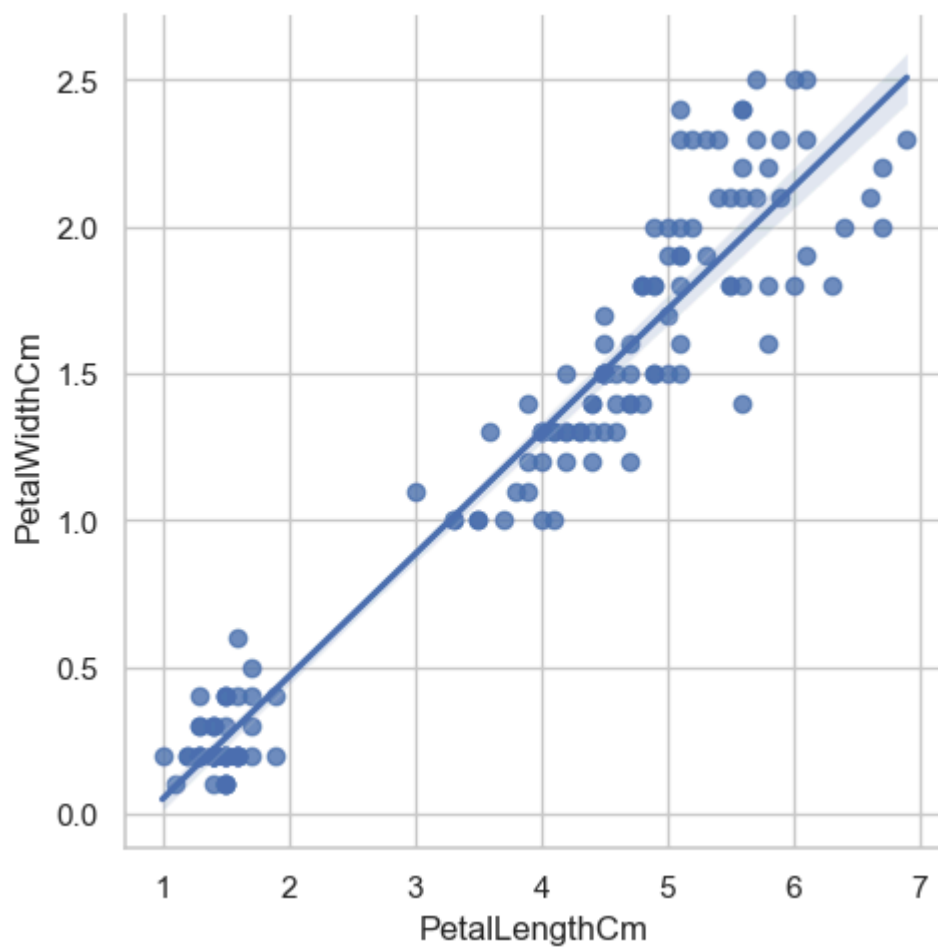
```
fig = sns.swarmplot(x="Species", y="PetalLengthCm", data=iris)
plt.show()
```



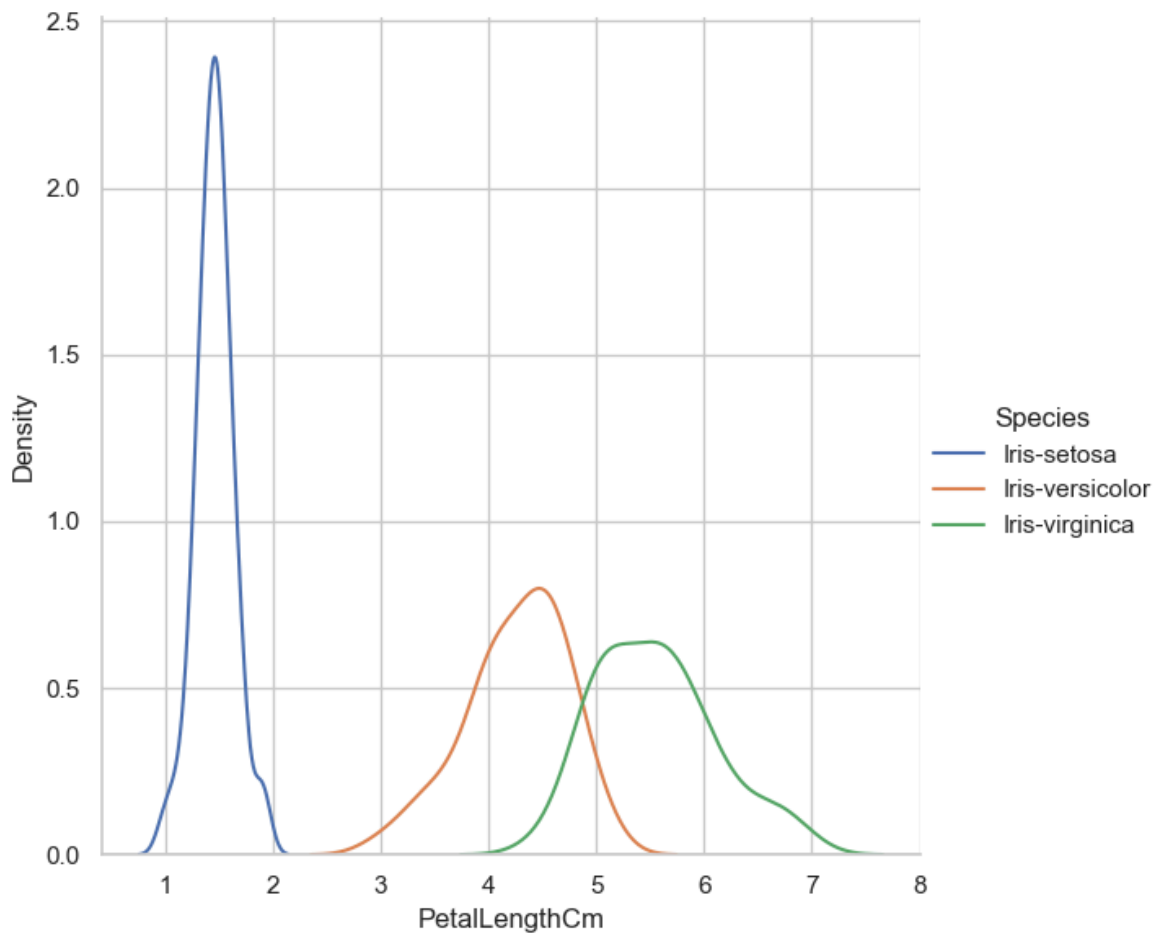
```
In [32]: sns.set(style="whitegrid")
fig=plt.gcf()
fig.set_size_inches(10,7)
ax = sns.violinplot(x="Species", y="PetalLengthCm", data=iris, inner=None)
ax = sns.swarmplot(x="Species", y="PetalLengthCm", data=iris,color="white", edge
plt.show()
```



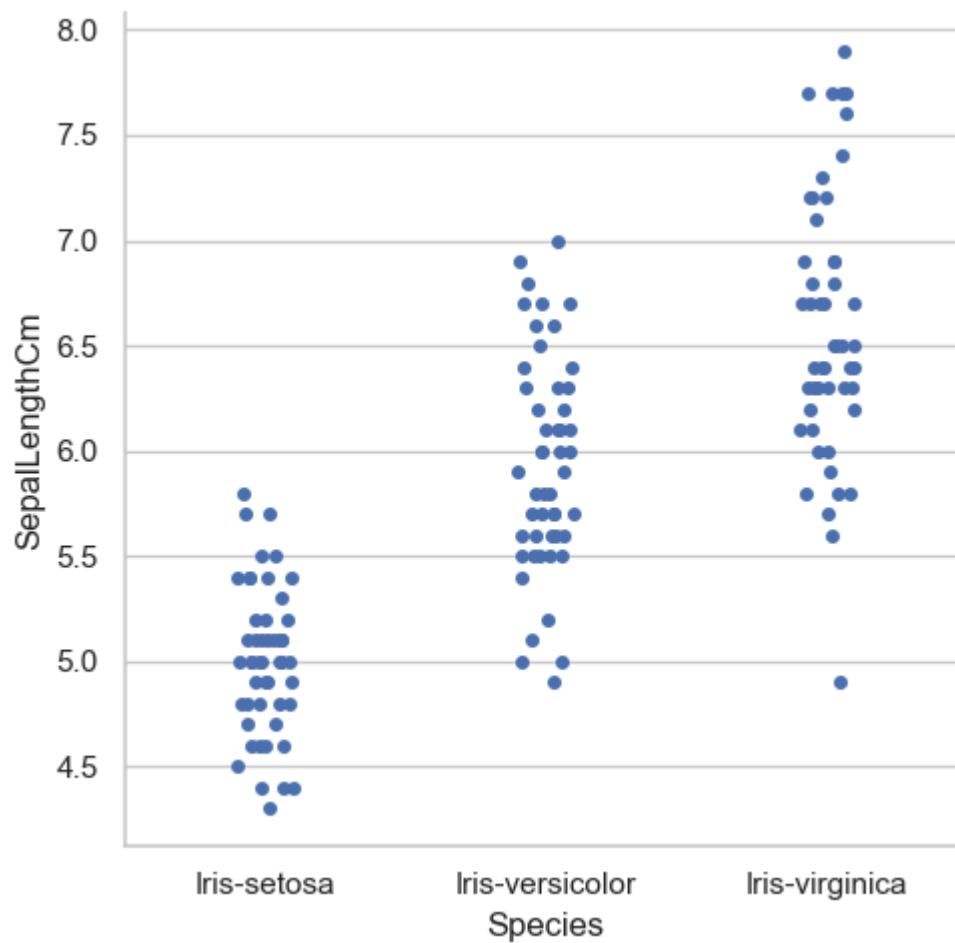
```
In [33]: fig=sns.lmplot(x="PetalLengthCm", y="PetalWidthCm",data=iris)
plt.show()
```



```
In [34]: g = sns.FacetGrid(iris, hue="Species", height=6)
g.map(sns.kdeplot, "PetalLengthCm").add_legend()
plt.show()
```



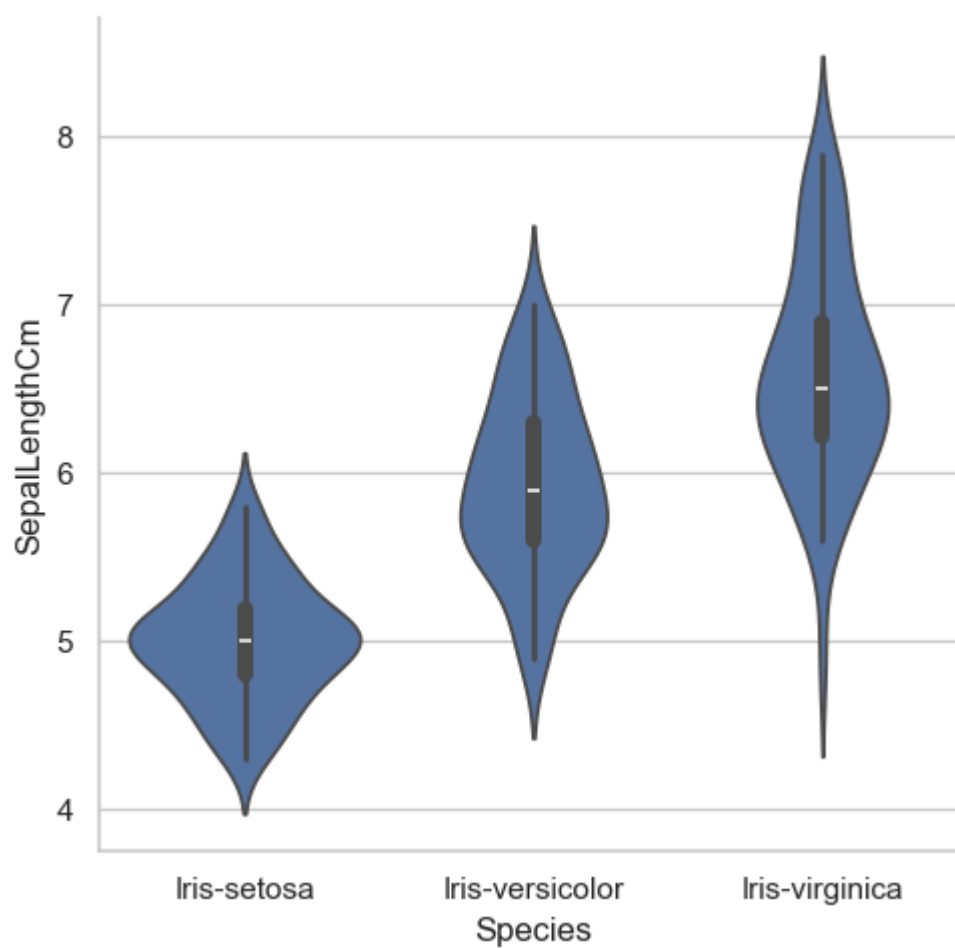
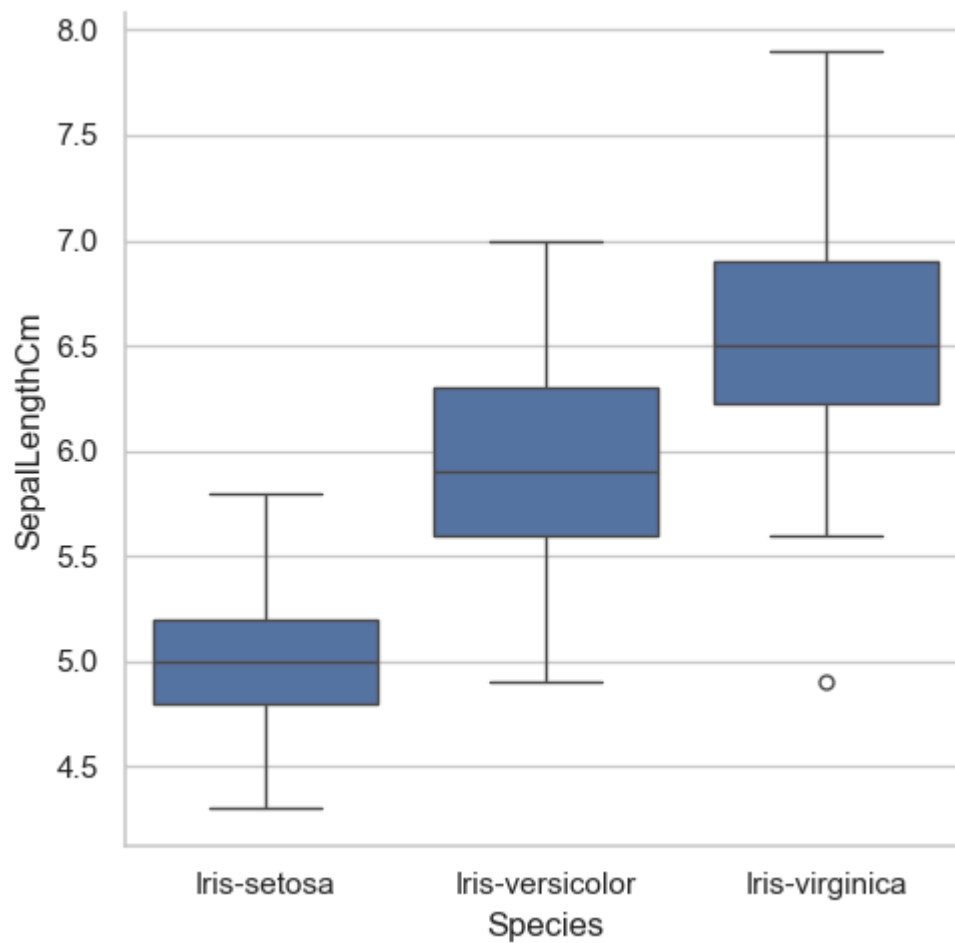
```
In [35]: sns.catplot(x='Species',y='SepalLengthCm',data=iris)
plt.ioff()
plt.show()
```



```
In [37]: # Box plot
sns.catplot(x='Species', y='SepalLengthCm', data=iris, kind='box')

# Violin plot
sns.catplot(x='Species', y='SepalLengthCm', data=iris, kind='violin')

plt.show()
```

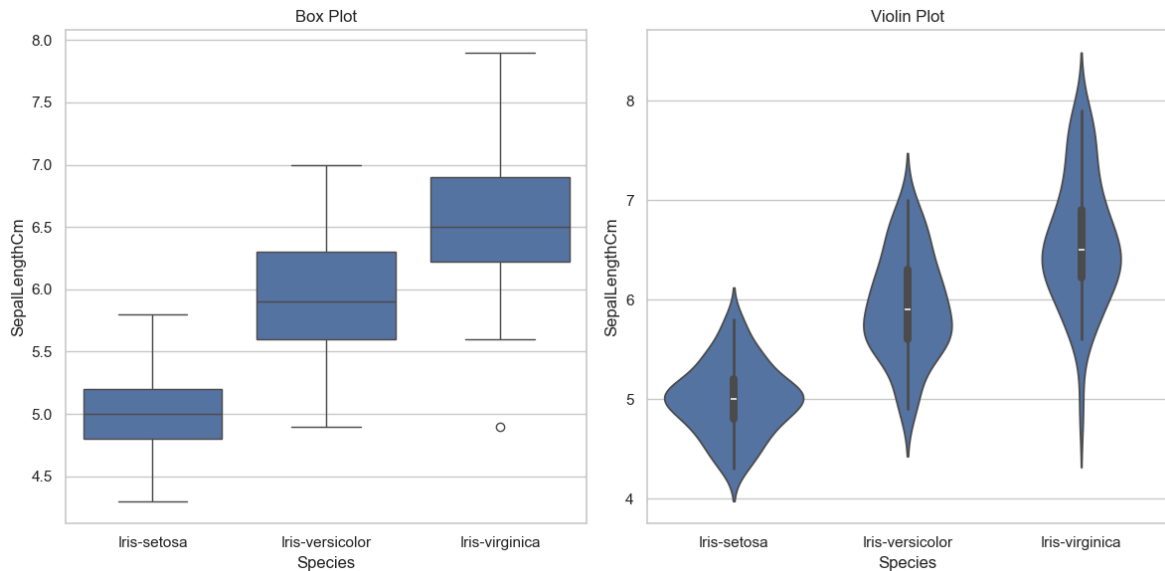


```
In [38]: fig, axes = plt.subplots(1, 2, figsize=(12, 6))
```

```
sns.boxplot(x='Species', y='SepalLengthCm', data=iris, ax=axes[0])
axes[0].set_title('Box Plot')

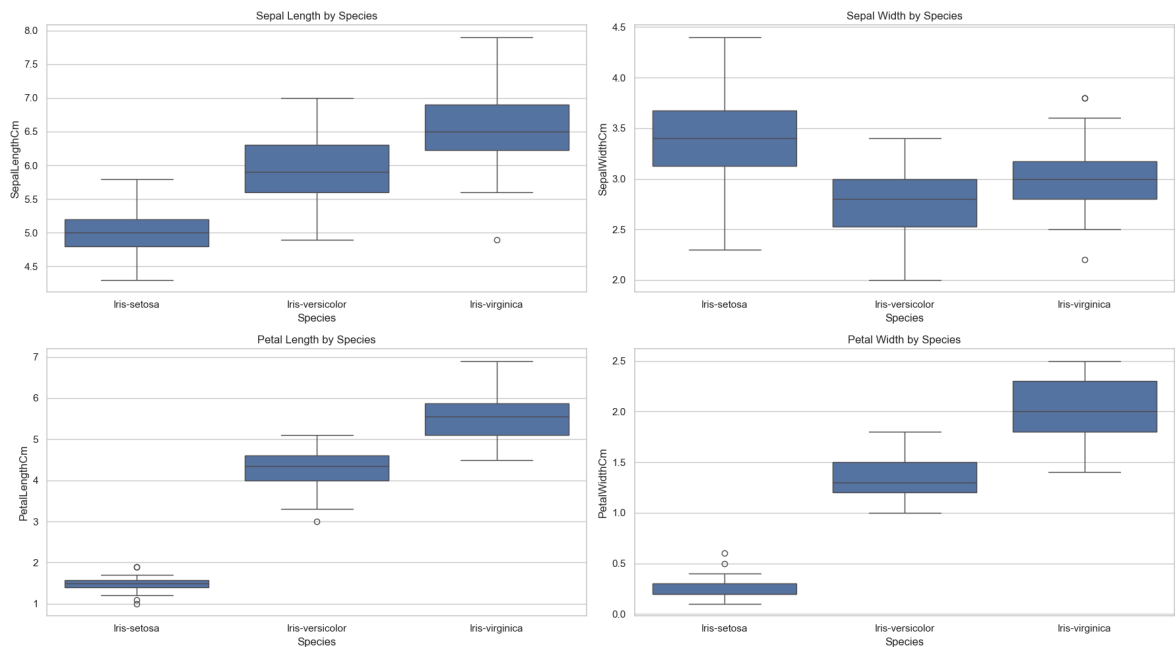
sns.violinplot(x='Species', y='SepalLengthCm', data=iris, ax=axes[1])
axes[1].set_title('Violin Plot')

plt.tight_layout()
plt.show()
```

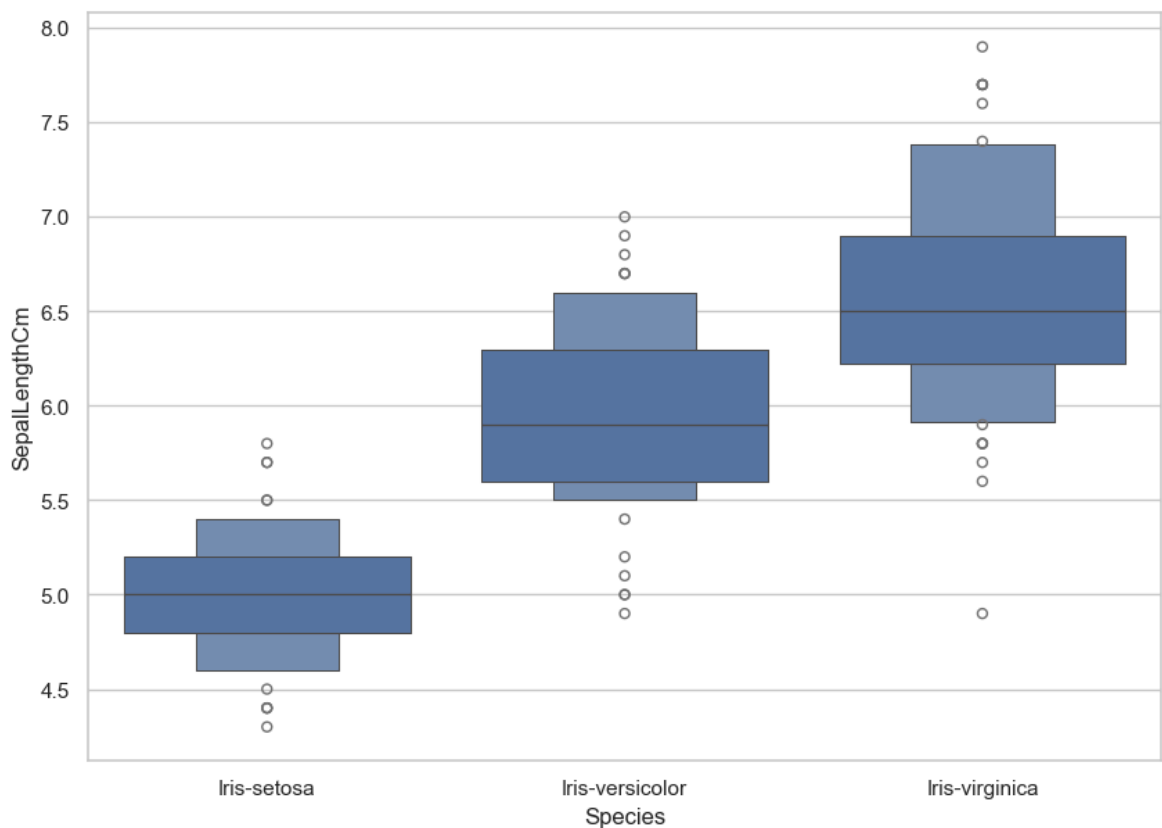


```
In [39]: import matplotlib.pyplot as plt
import seaborn as sns
```

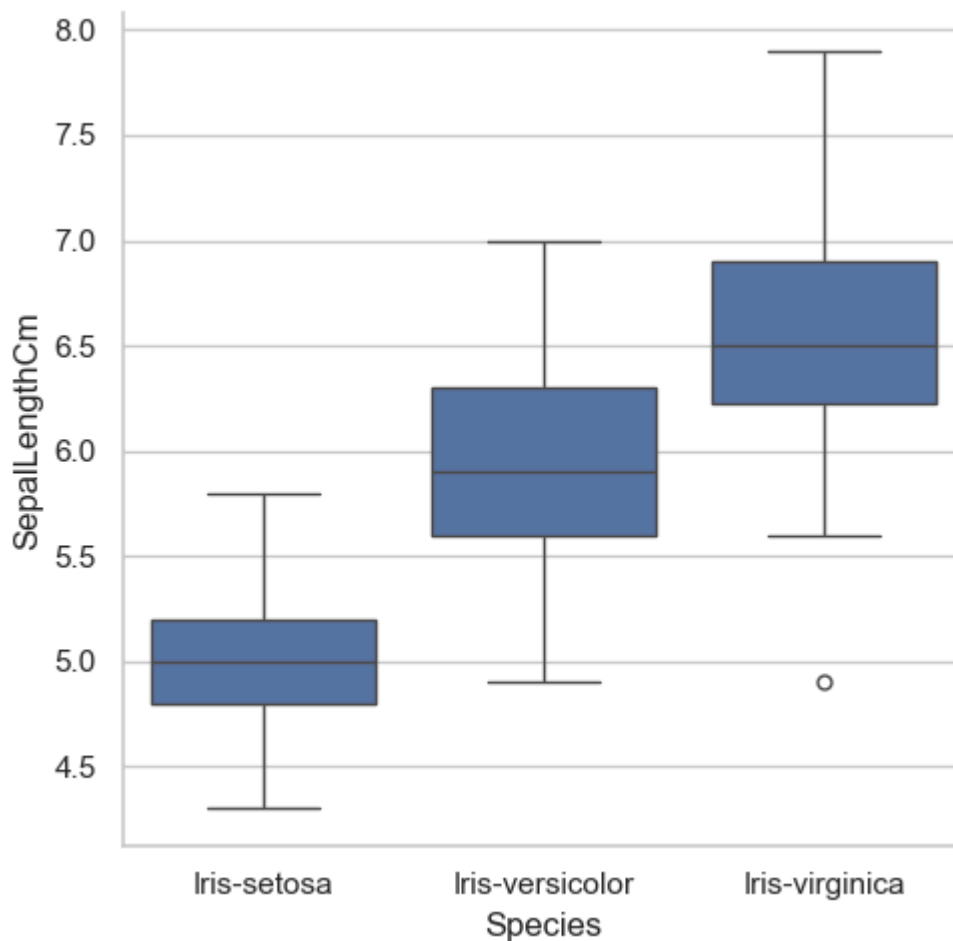
```
In [40]: # Create a 2x2 grid of subplots
fig, ax = plt.subplots(2, 2, figsize=(18, 10))
# Plotting each feature using Seaborn boxplot
sns.boxplot(x='Species', y='SepalLengthCm', data=iris, ax=ax[0][0])
sns.boxplot(x='Species', y='SepalWidthCm', data=iris, ax=ax[0][1])
sns.boxplot(x='Species', y='PetalLengthCm', data=iris, ax=ax[1][0])
sns.boxplot(x='Species', y='PetalWidthCm', data=iris, ax=ax[1][1])
# Optional: Add titles to each subplot
ax[0][0].set_title('Sepal Length by Species')
ax[0][1].set_title('Sepal Width by Species')
ax[1][0].set_title('Petal Length by Species')
ax[1][1].set_title('Petal Width by Species')
plt.tight_layout()
plt.show()
```



```
In [41]: fig=plt.gcf()
fig.set_size_inches(10,7)
fig=sns.boxenplot(x='Species',y='SepalLengthCm',data=iris)
plt.show()
```



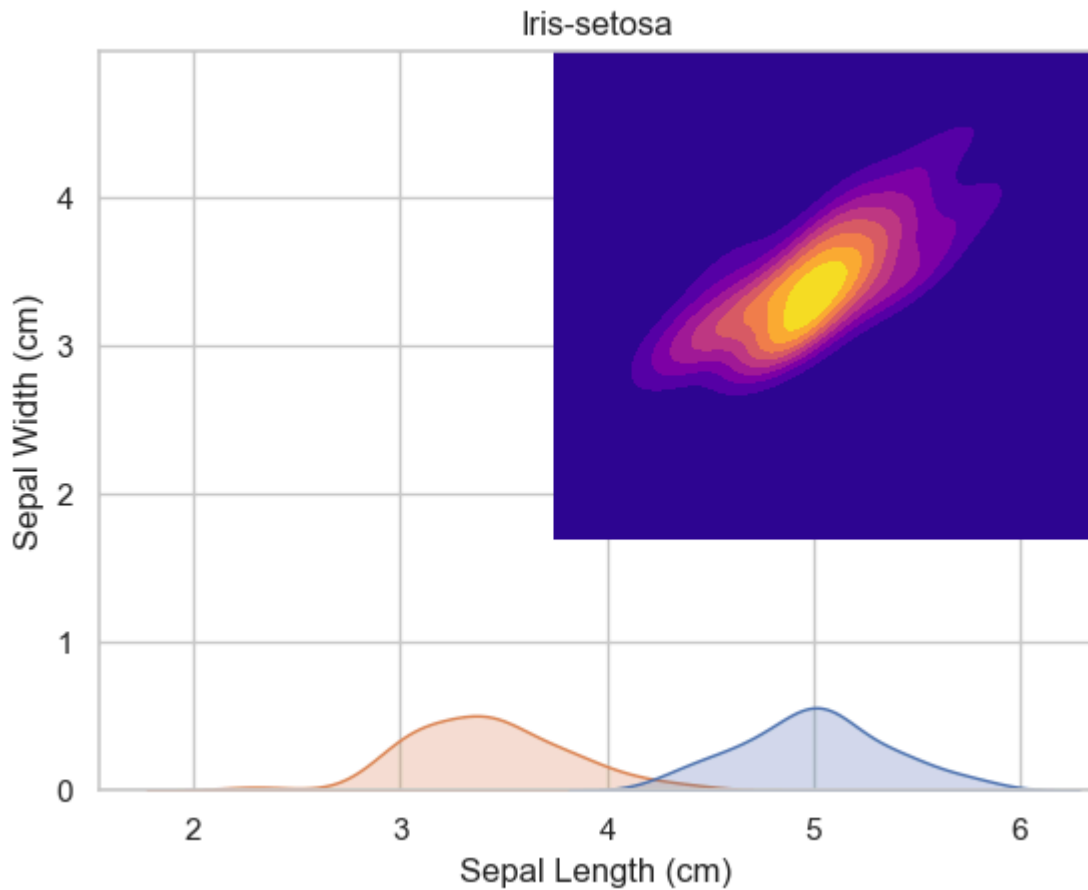
```
In [42]: sns.catplot(x='Species', y='SepalLengthCm', data=iris, kind='box')
plt.show()
```



```
In [44]: # Filter for Iris-setosa species
sub = iris[iris['Species'] == 'Iris-setosa']

# Create 2D KDE plot
sns.kdeplot(
    x=sub['SepalLengthCm'],
    y=sub['SepalWidthCm'],
    cmap="plasma",
    fill=True, # replaces shade=True
    thresh=0   # ensures full shading
)

# Plot Labels and title
plt.title('Iris-setosa')
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Sepal Width (cm)')
plt.show()
```



```
In [45]: sns.set_style('darkgrid')
f, axes = plt.subplots(2, 2, figsize=(15, 15))

# Boxplot
sns.boxplot(x="Species", y="PetalLengthCm", data=iris, ax=axes[0, 0])

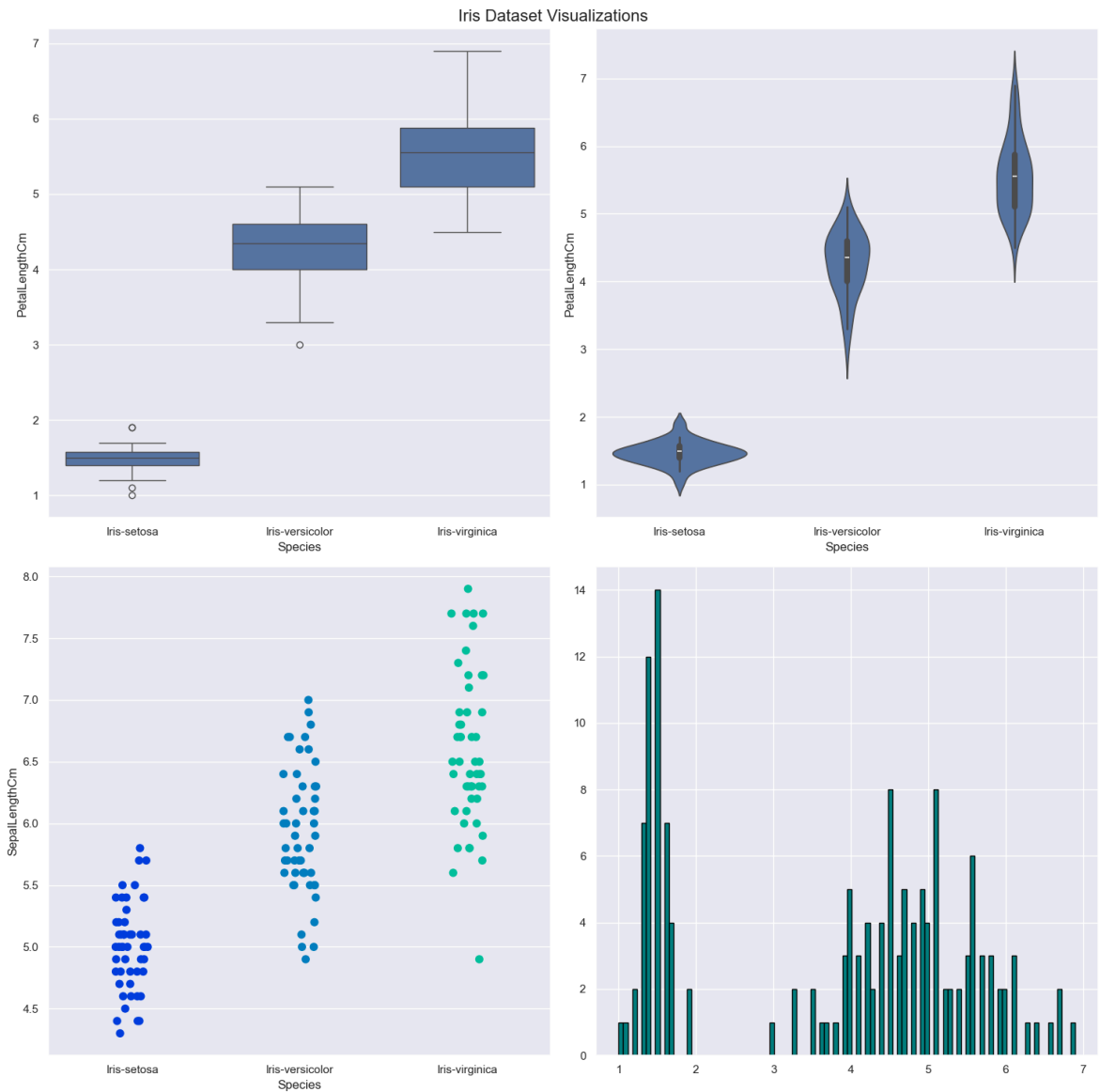
# Violin plot
sns.violinplot(x='Species', y='PetalLengthCm', data=iris, ax=axes[0, 1])

# Strip plot
sns.stripplot(x='Species', y='SepalLengthCm', data=iris,
              jitter=True, edgecolor='gray', size=8,
              palette='winter', orient='v', ax=axes[1, 0])

# Histogram - corrected this line
axes[1, 1].hist(iris['PetalLengthCm'], bins=100, color='teal', edgecolor='black')

# Optional: Set a title for the whole figure
f.suptitle('Iris Dataset Visualizations', fontsize=16)

plt.tight_layout()
plt.show()
```



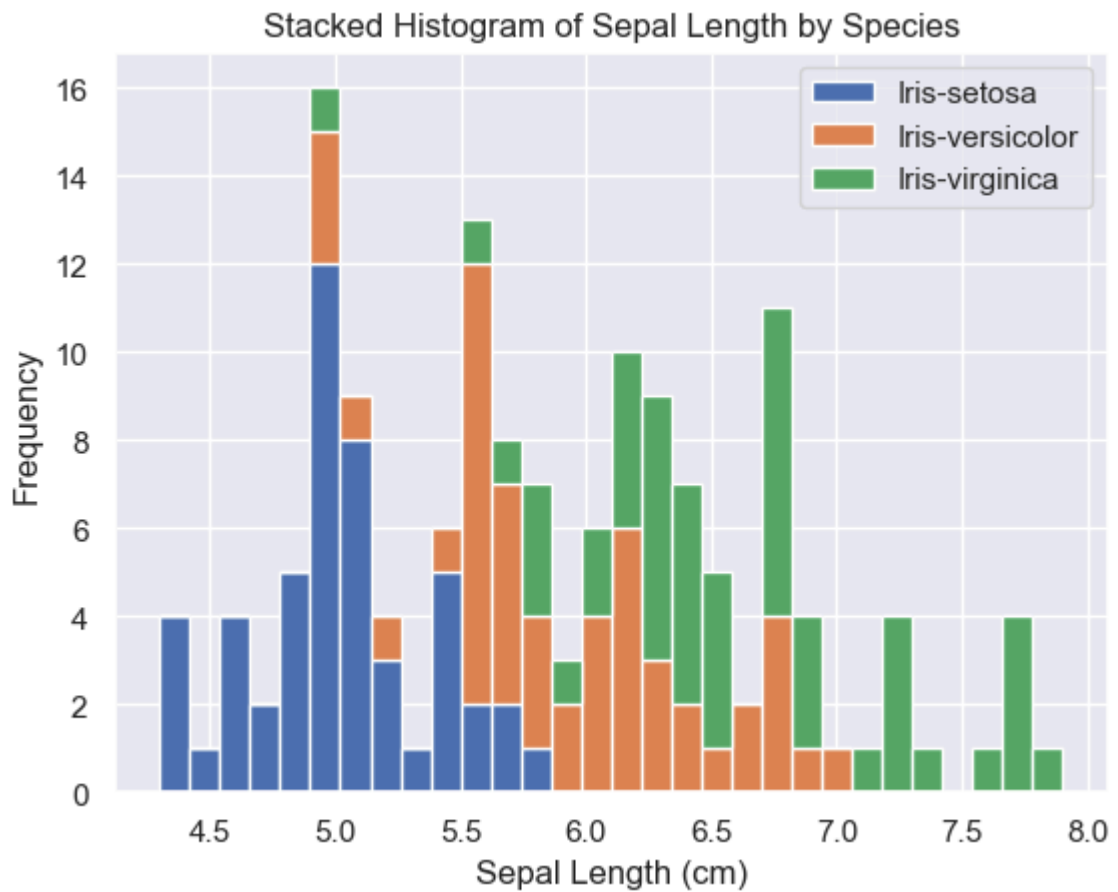
```
In [46]: iris['Species'] = iris['Species'].astype('category')
#iris.head()
```

```
In [47]: iris['Species'] = iris['Species'].astype('category')

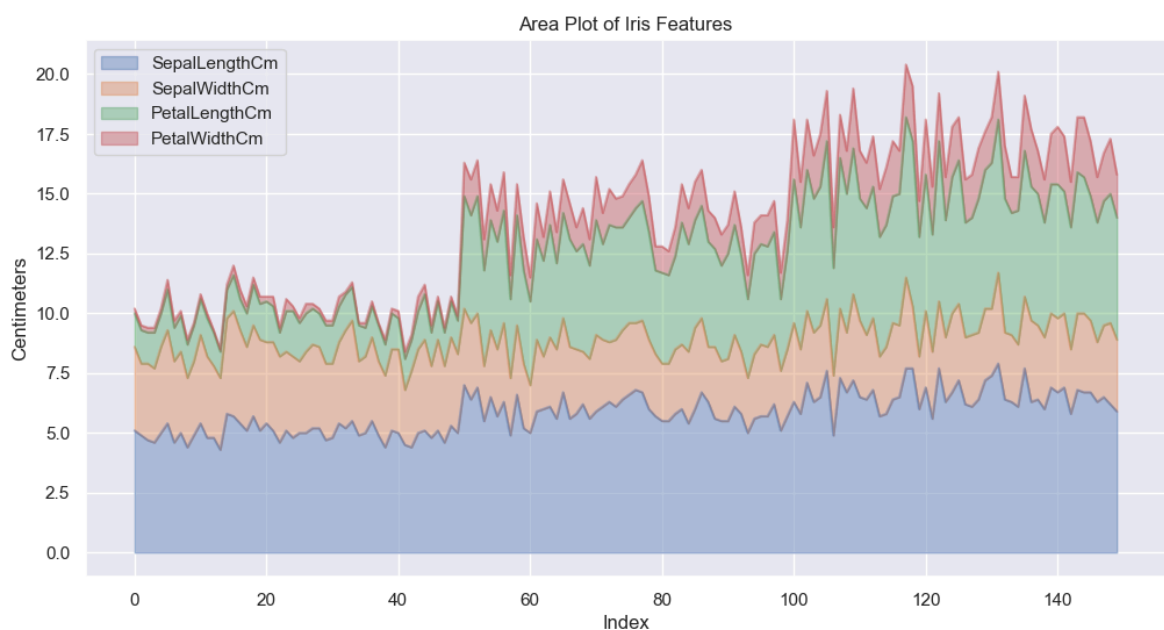
list1 = []
mylabels = []

# Grouping data by species
for gen in iris['Species'].cat.categories:
    list1.append(iris[iris['Species'] == gen]['SepalLengthCm'])
    mylabels.append(gen)

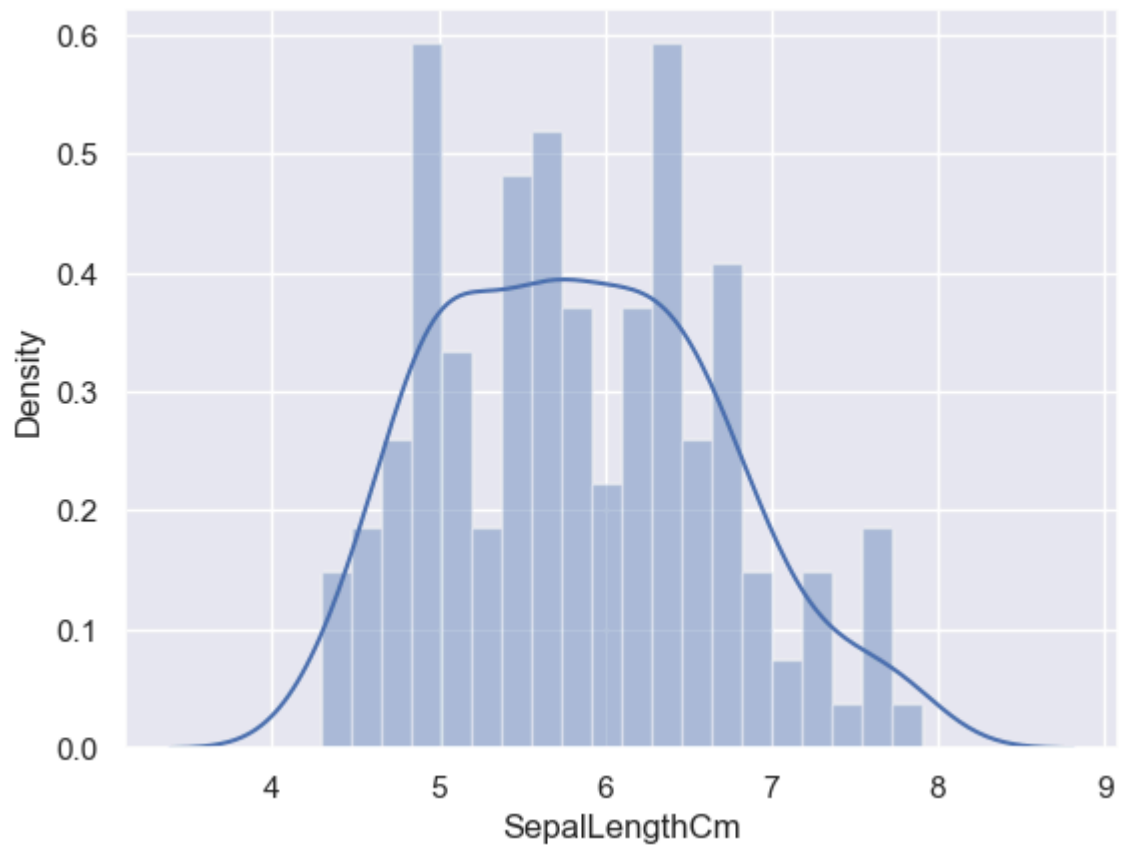
# Plot stacked histogram
plt.hist(list1, bins=30, stacked=True, rwidth=1, label=mylabels)
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Frequency')
plt.title('Stacked Histogram of Sepal Length by Species')
plt.legend()
plt.show()
```



```
In [48]: iris.plot.area(  
    y=['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm'],  
    alpha=0.4,  
    figsize=(12, 6)  
)  
  
plt.title("Area Plot of Iris Features")  
plt.ylabel("Centimeters")  
plt.xlabel("Index")  
plt.show()
```



```
In [49]: sns.distplot(iris['SepalLengthCm'], kde=True, bins=20);  
plt.show()
```



```
In [ ]:
```